Callum Battilana

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Skills

Languages. **Proficient:** Python, R
Experienced: SQL, Excel

Tech Stack Jupyter Notebook (Anaconda), Pycharm, Rstudio, Git, Microsoft Copilot

Experience

Bureau of Meteorology | Data Scientist Intern — **Jul 2025 – Oct 2025**, *Docklands, Melbourne*

- Developed an AI Convolutional Neural Network machine learning model for the challenging problem of estimating global wind patterns every hour across the globe using Python. The technical tools used include Pandas, Numpy, Sci-kit Learn, PyTorch, and Tensorflow.
- **The model's accuracy was 78 percent on average across all regions of the globe, while only receiving data from just 0.7 percent of the globe**. This model could **increase forecasting efficiency by 100 times** compared to traditional approaches. **Global forecasts of wind patterns hours ahead can be generated in a few minutes instead of hours**.
- Presented my findings globally to climate researchers who were part of the APARC Reanalysis Inter-Comparison Project. Wrote a 13,000 word thesis which recieved a high distinction. Performed intensive research on the latest deep-learning methods in climate science. Findings will be published soon in an academic journal.

Asahi (Carlton United Breweries) | Data Scientist Intern — **Jul 2024 – Oct 2024**, *Abbotsford, Melbourne*

- Used Excel VBA to develop prototypes of various Statistical Process Control (SPC) charts that can predict future issues with can seamer quality in real time. Found 3 new charts that the company did not consider implementing before. Preliminary tests to validate chart performance was done using R.
- These charts **were able to diagnose problems weeks before they were found manually by the company.** If these charts were used earlier by the company, **5 million dollars would have been saved** in 2023 alone.
- These charts apply forecasting techniques to analyse trends in manufacturing quality such as taking the moving average. Statistical tests on these charts such as the ANOVA and t-test are then used to highlight future issues with can seamer quality.
- Worked in a cross-functional team with Engineers, Mechanical Operators, and Quality Maintenance professionals.

Education

Monash University — **Feb 2022 – Nov 2025**, *Clayton, Melbourne*
Applied Data Science (Honours): ***Grade: First Class Honours (GPA Overall: 3.281/4)***

Hobbies

- Write Data Science related articles on Medium.
- Keen chess player. Played in competitions at school, and I still play frequently online.
- Play viola: Joined an orchestra at University.

Volunteering

Interchange Outer East — **Mar 2025 – Present**, *Ferntree Gully, Melbourne*

- Looking after disabled people and socializing with them while going on fun activities.